



COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5701

Probability

Dobrin Marchev

Deterministic Systems

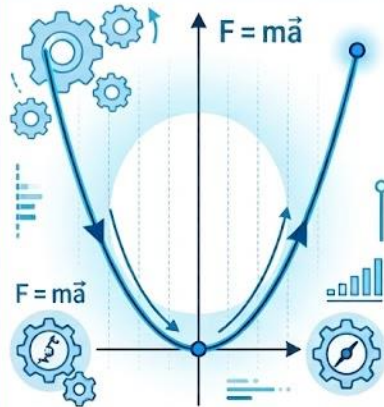
- **Mathematical Formulation:** Environments defined by a closed-form analytic mapping where a given feature space $\mathbf{x}$ guarantees a singular, absolute outcome y . This is represented as the strict functional relationship $y = f(\mathbf{x})$.
- **Error Profile:** The irreducible error (aleatoric uncertainty) in the system is strictly zero ($\text{Var}(\epsilon) = 0$). Any deviation between a prediction and the outcome is purely epistemic—stemming from measurement noise or selecting the wrong functional form, rather than randomness in the phenomenon itself.
- **Domain Context:** Prevalent in theoretical physics, formal logic, and computer science mechanics where all latent variables are observable and strictly governed by established laws.
- **Applied Example:** Rule-based feature scaling or deterministic unit transformations in a data pipeline, such as absolute temperature conversion:
$$T_{\text{Celsius}} = (T_{\text{Fahrenheit}} - 32) \times \frac{5}{9}.$$

Stochastic (Non-Deterministic) Processes

- **Mathematical Formulation:** Environments where the target variable Y is a random variable mapped to a probability distribution conditioned on the inputs $\mathbf{x}$. This is modeled as $y = f(\mathbf{x}) + \epsilon$, where ϵ denotes intrinsic noise.
- **Error Profile:** The system is bounded by irreducible error ($\text{Var}(\epsilon) > 0$) caused by unobserved confounding variables, chaotic dynamics, or macro-level entropy. Consequently, achieving a "perfect" point prediction is mathematically impossible, regardless of algorithmic complexity or compute power.
- **Domain Context:** The foundational paradigm of data science and machine learning. Because absolute certainty is impossible, models are designed to optimize and approximate the conditional expectation $\mathbb{E}[Y|\mathbf{x}]$, likelihood estimations, or decision boundaries within a specified confidence threshold.
- **Applied Example:** Predicting user churn, macroeconomic forecasting, or natural language generation; the underlying phenomena are inherently probabilistic, and model outputs represent a distribution of likely states rather than an exact physical guarantee.

Phenomena

DETERMINISTIC SYSTEM



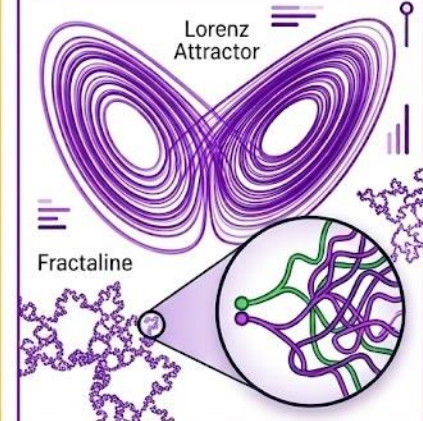
PRECISE, RULE-BASED, AND PERFECTLY PREDICTABLE. THE FUTURE IS DEFINED BY THE PRESENT STATE.

STOCHASTIC SYSTEM



PROBABILISTIC AND NON-DETERMINISTIC. OUTCOMES ARE SUBJECT TO CHANCE & RANDOM NOISE.

UNPREDICTABLE CHAOS



EXTREMELY SENSITIVE TO INITIAL CONDITIONS. A SMALL CHANGE LEADS TO VASTLY DIFFERENT, NON-PERIODIC OUTCOMES.

Random phenomena

- Unable to predict the outcomes, but in the long-run, the outcomes exhibit statistical regularity.
- The probability does *not* depend on interpretation.

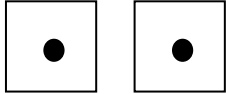
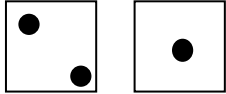
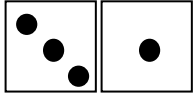
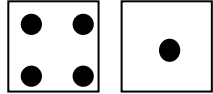
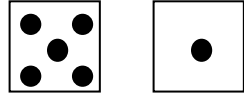
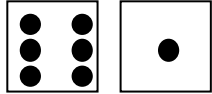
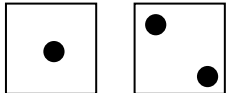
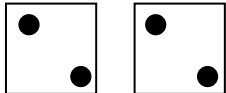
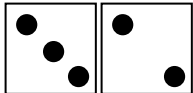
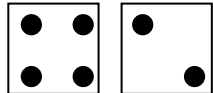
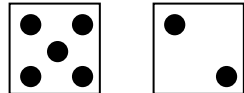
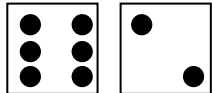
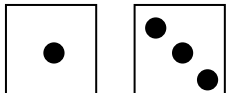
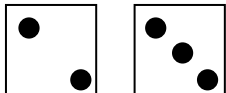
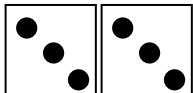
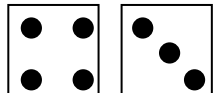
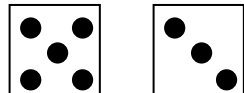
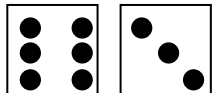
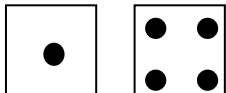
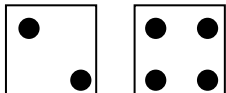
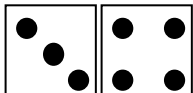
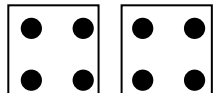
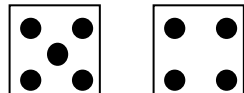
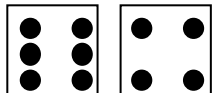
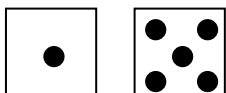
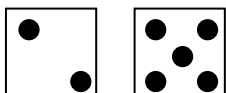
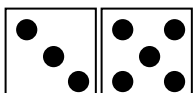
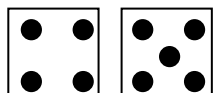
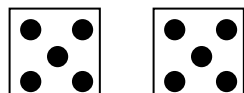
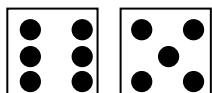
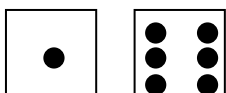
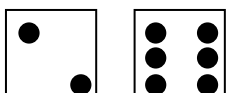
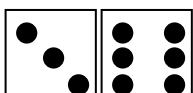
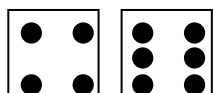
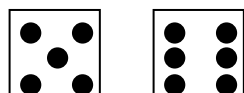
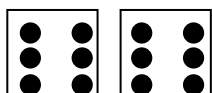
Examples

- Tossing a coin – outcomes $S = \{\mathbf{Head}, \mathbf{Tail}\}$

Unable to predict on each toss whether is Head or Tail.

In the long run can predict that 50% of the time heads will occur and 50% of the time tails will occur

- Rolling two balanced dice – 36 outcomes

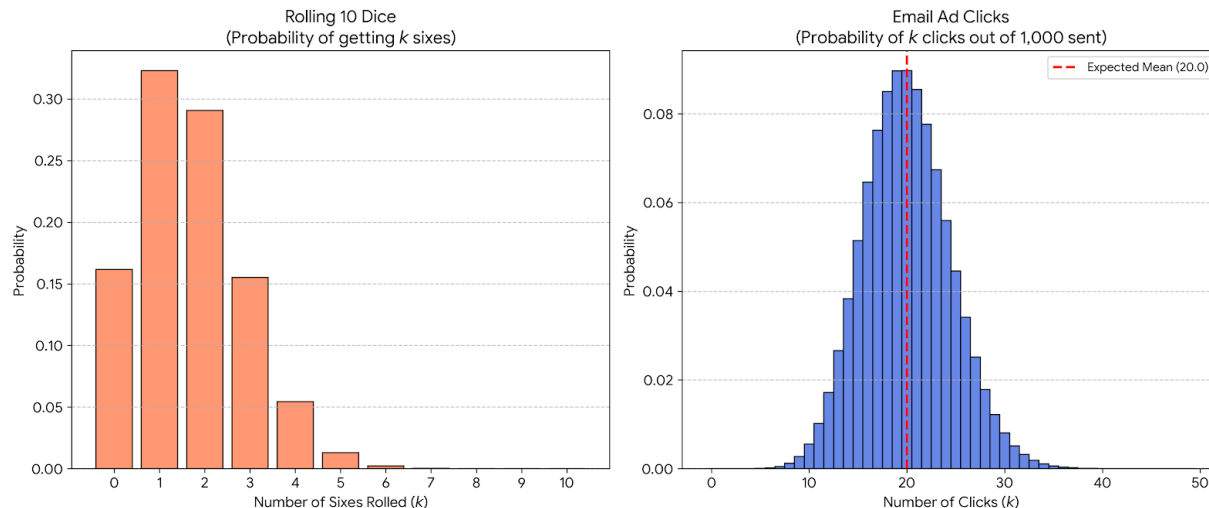
					
					
					
					
					
					

Rolling Multiple Dice -> Website Traffic (Binomial Distribution)

The setup: You roll 10 dice at once and count how many times a 6 appears.

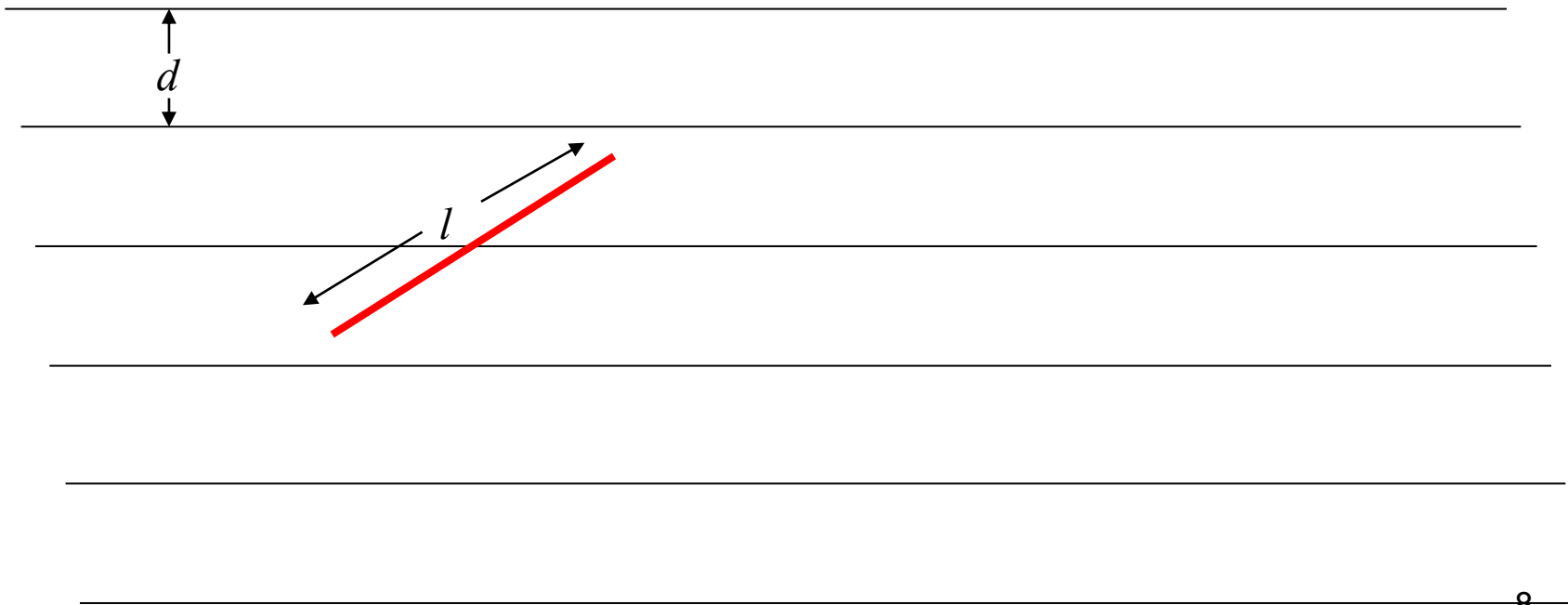
Data science use: This is a **Binomial distribution**, tracking the total number of successes in a set number of independent yes/no trials.

Real-world use: Predicting the number of people who will click an email ad out of 1,000 sent emails.

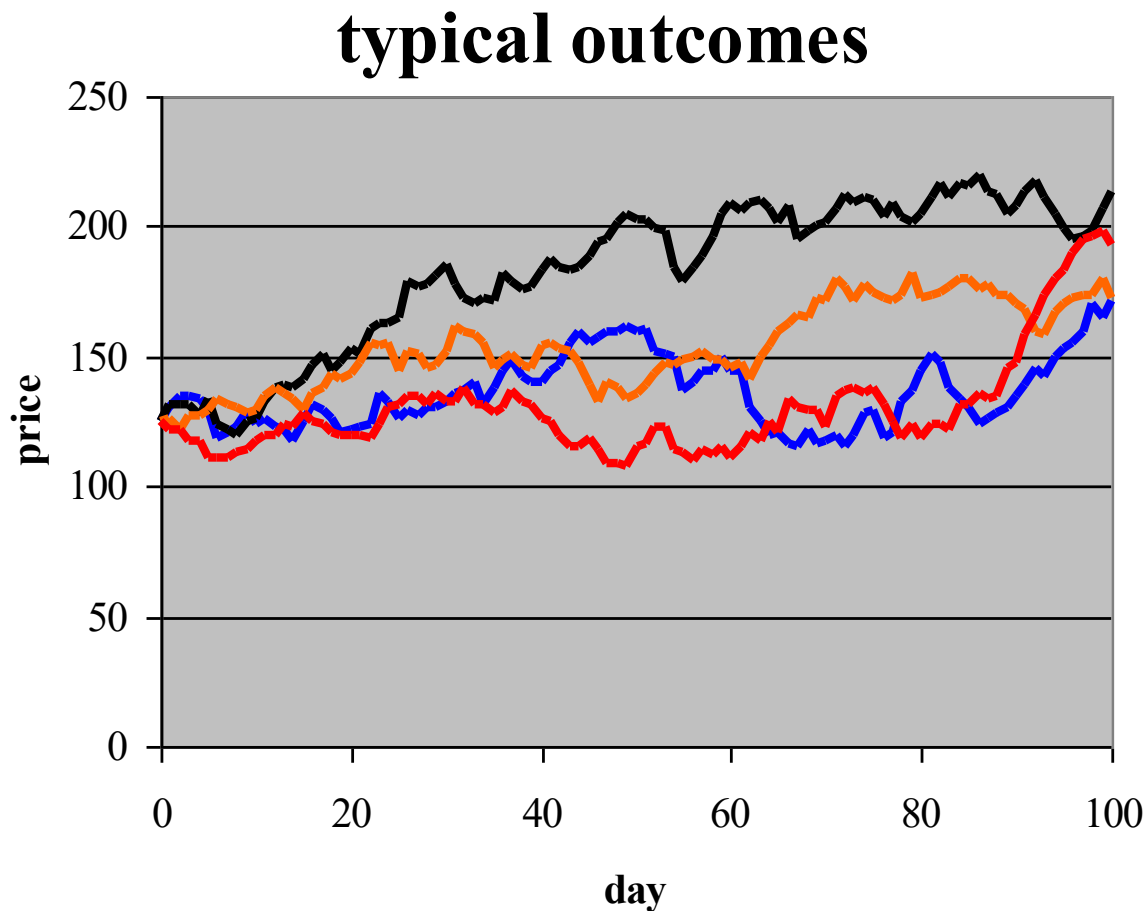


- Buffon's Needle problem (18th century)
 - A needle of length l , is tossed and allowed to land on a plane that is ruled with horizontal lines a distance, d , apart

A typical outcome

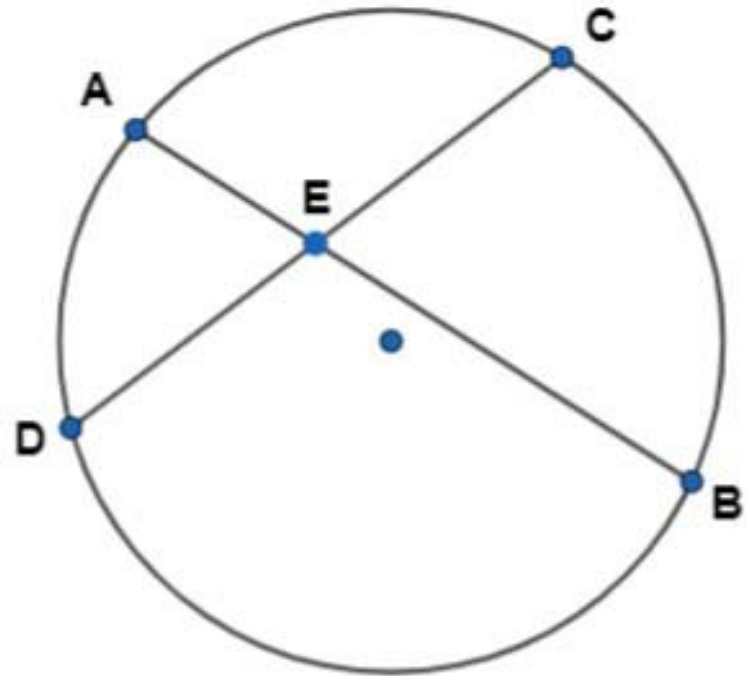


- Stock market performance (20th century)
 - A stock currently has a price of \$125.50. We will observe the price for the next 100 days



- Data Science interview question

What is the probability that two random chords AB and CD intersect at a point inside the circle?



Definitions

Sample Space

The **sample space**, S , of an experiment is the set of *all possible different* outcomes.

The sample space S may contain:

1. A finite number of outcomes.
2. A countably infinite number of outcomes, or
3. An uncountably infinite number of outcomes.

Sample Space

- A sample space S is **countably infinite** if there exists a bijective mapping (a one-to-one and onto correspondence) between the elements of S and the set of positive integers $\mathbb{Z}^+$ (or natural numbers $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$).
- This bijection guarantees that the elements of S can be sequentially indexed as a countably infinite set of discrete outcomes:

$$S = \{O_i \mid i \in \mathbb{Z}^+\}$$

where O_i represents the i -th distinct outcome.

- **Distribution Modeling:** It forms the theoretical basis for modeling unbounded count data, enabling the use of models like the Poisson, Geometric, and Negative Binomial distributions for tasks such as queuing theory, website traffic prediction, and failure rate analysis.

- Rolling a two balanced dice – 36 outcomes
The sample space S for rolling two standard six-sided dice consists of 36 discrete outcomes, represented as ordered pairs (D_1, D_2) where each variable represents the face value of a die.

$$\Omega = \{(i, j) \mid i, j \in \{1, 2, 3, 4, 5, 6\}\}$$

- Evaluating a hyperparameter grid search for a machine learning model across multiple discrete parameters yields a finite sample space of 24 distinct training outcomes.

Grid Search Configuration

n_estimators: {100, 200, 300, 400} (4 discrete states)

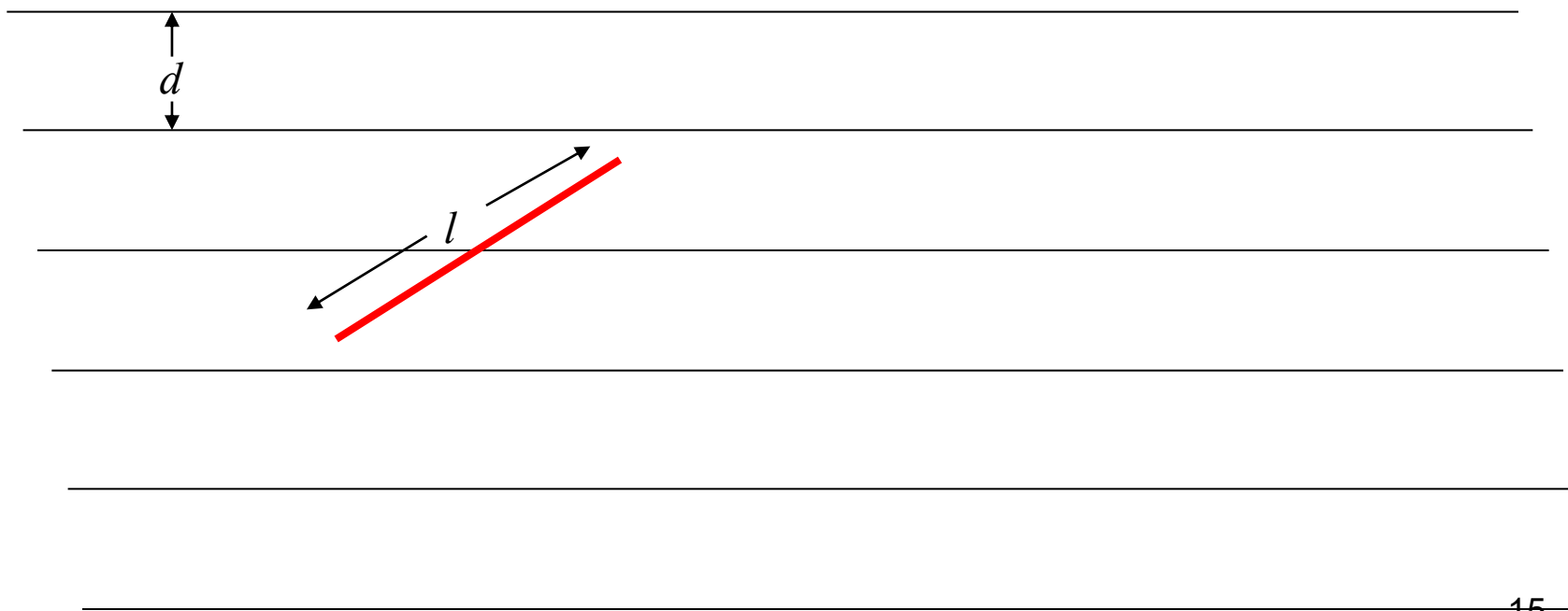
max_depth: {10, 20, 30} (3 discrete states)

criterion: {'gini', 'entropy'} (2 discrete states)

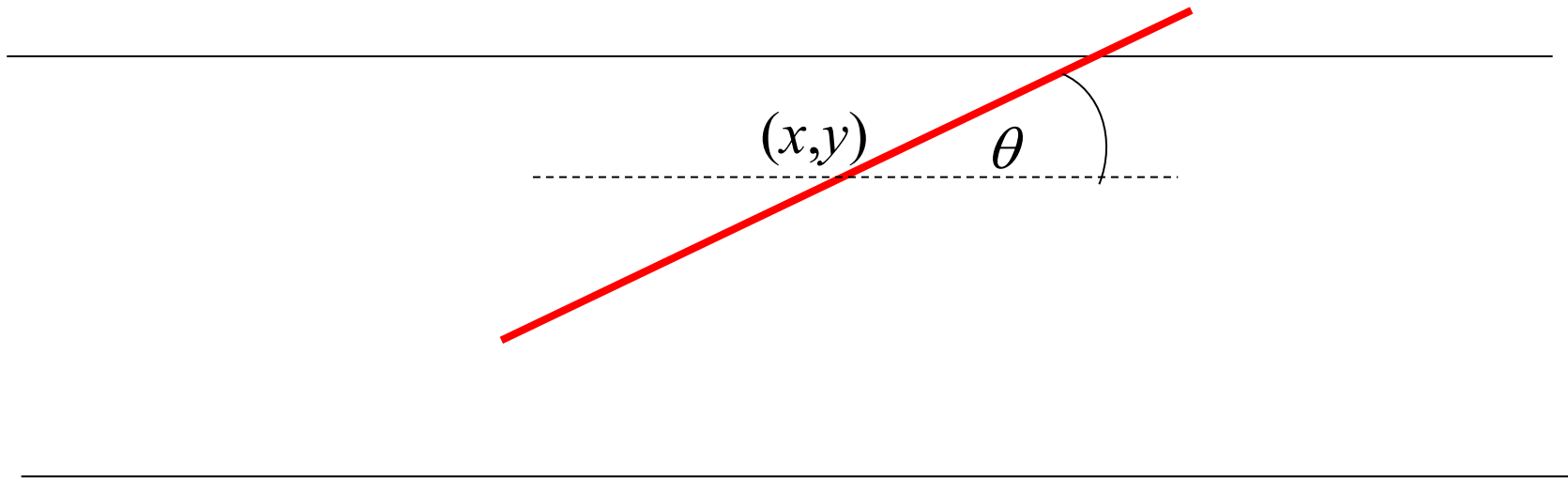
Time to think: What is the sample space S ?

- Buffon's Needle problem
 - A needle of length l , is tossed and allowed to land on a plane that is ruled with horizontal lines a distance, d , apart

A typical outcome



An outcome can be identified by determining the coordinates (x, y) of the center of the needle and θ , the angle the needle makes with the parallel ruled lines.



$$S = \{(x, y, \theta) \mid -\infty < x < \infty, 0 < y < d, 0 \leq \theta \leq 2\pi\}$$

Alternatives:

- Midpoint perpendicular distance to nearest line:

$x \sim \text{Uniform}(0, d/2)$ and
orientation angle $\theta \sim \text{Uniform}(0, \pi/2)$.

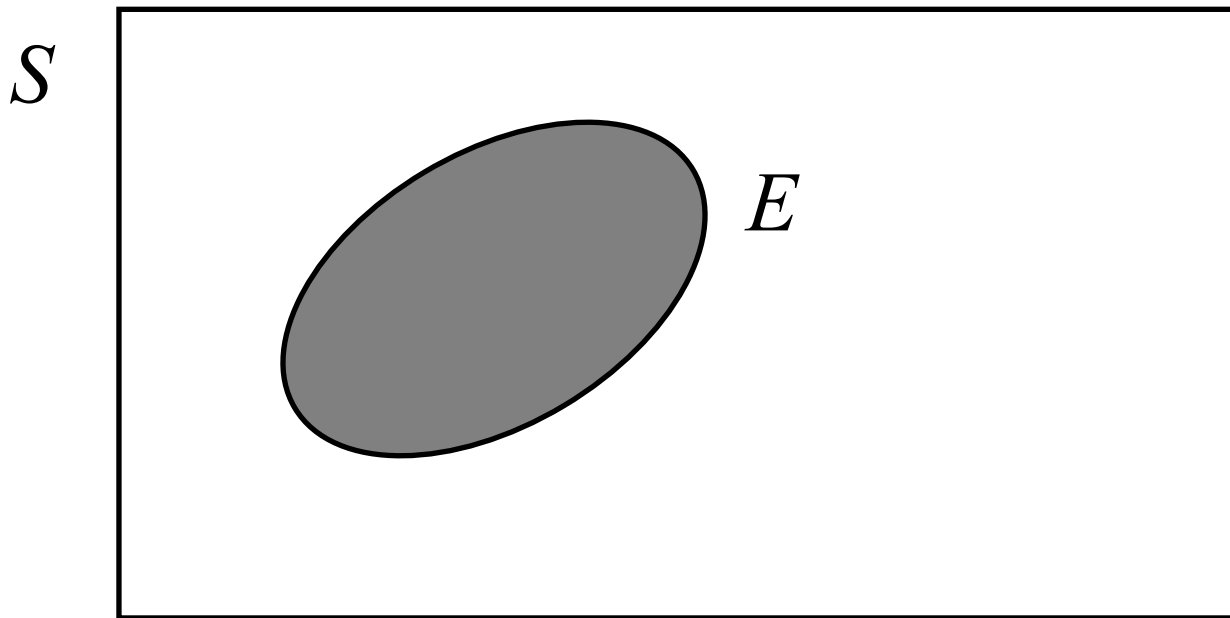
- **Buffon-Laplace 2D Grid Parametrization
(Spatial Statistics)**

Distances $x \in [0, a/2]$ and $y \in [0, b/2]$ to nearest vertical and horizontal lines spaced a and b apart, with angle $\theta \in [0, \pi/2]$ for $l < \min(a, b)$.

Time to think: Do you know other similar problems?

Events

The **event**, E , is any subset of the **sample space**, S .



More Set Notation

- Given a set A (which may represent a sample space, an event, or a feature domain), the proposition $e \in A$ asserts that the distinct entity e is a constituent member of A .
- Conversely, $e \notin A$ denotes that the element is strictly excluded from the set (does not belong)

Applications in Data Science

- **Probability Theory:** It specifies that a discrete experimental outcome e belongs to a specific event subset A within the global sample space S
- **Feature Engineering:** It formalizes categorical assignments, such as mathematically defining whether a specific observation's feature value x_i belongs to a discrete target class C_k ($x_i \in C_k$).

Example 1

Three cars at an intersection can turn either left (L) or right (R).

a) Define the sample space S

b) Determine the outcomes of the following events:

A = “Exactly one car turns right”

C = “All cars turn in the same direction”

Special Events

- The null event is a subset of the sample space containing no experimental outcomes, denoted as $\emptyset = \{\}$
- **Later:** By Kolmogorov's axioms, the probability measure of the empty event is strictly zero: $P(\emptyset) = 0$.
- **The Entire Event, the Sample Space – S**

S = the event that contains all outcomes

- The empty event, $\emptyset$, never occurs.
- The entire event, S , always occurs.

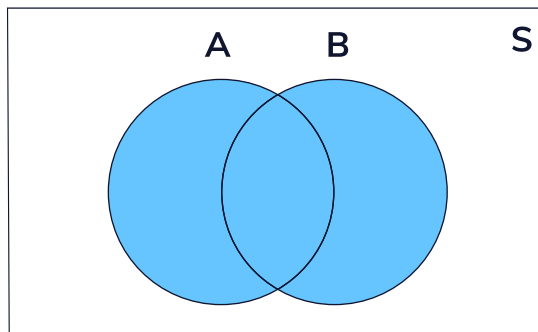
Some applications:

- **Query Operations:** In relational databases or dataframe manipulations, returning a null event corresponds to an empty result set (e.g., an empty Pandas DataFrame or an SQL return of zero rows) when evaluating disjoint filter conditions.

Set operations on events

- Given a sample space S , the **union** of two events A, B , denoted $A \cup B$, is the set of all elementary outcomes e that belong to A , to B , or to both:

$$A \cup B = \{e \in S \mid e \in A \text{ or } e \in B\}$$



- For an arbitrary indexed family of events $\{A_i\}_{i \in I}$ (where I can be finite, countably infinite, or uncountably infinite), the generalized union is defined as:

$$\bigcup_{i \in I} A_i = \{e \in S \mid \exists i \in I \text{ such that } e \in A_i\}$$

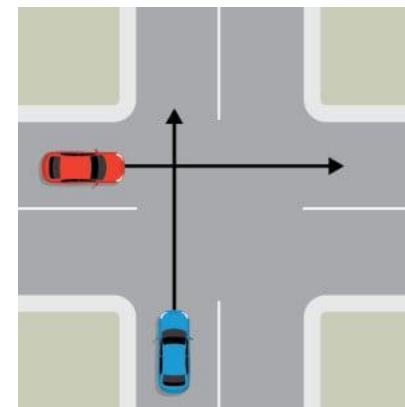
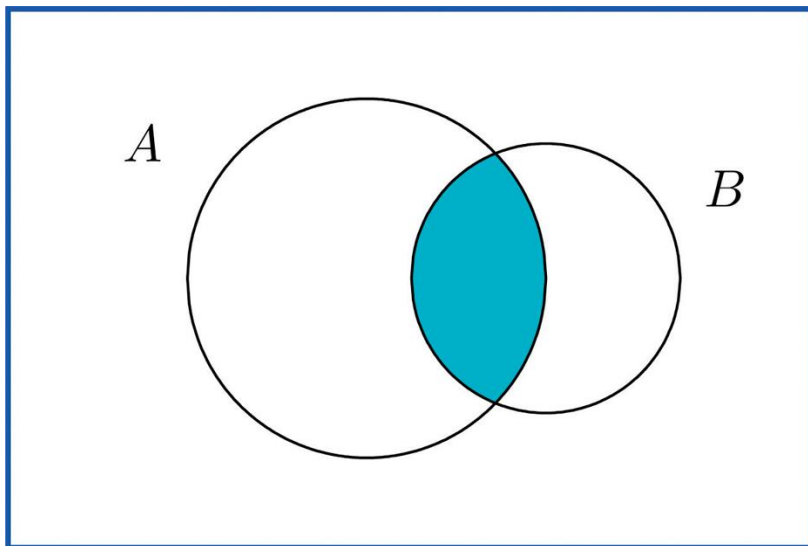
Intersection

Given a sample space S and events $A, B \subset S$, the intersection is denoted mathematically as:

$$A \cap B = \{e \in S \mid e \in A \text{ and } e \in B\}$$

Applications:

Joint Probability: The intersection evaluates the joint probability mass or density, $P(A \cap B)$, modeling the exact likelihood of two features, states, or labels co-occurring in a given dataset.



Exercise 1

Sketch the region in the xy -coordinate plane corresponding to $A \cup B$ and $A \cap B$ if

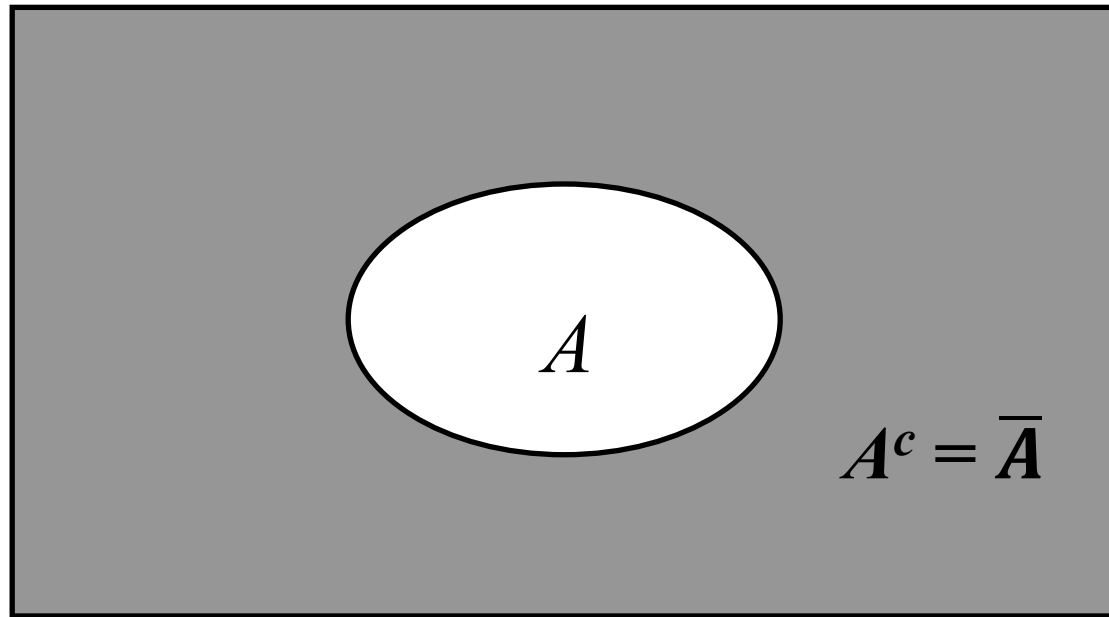
$$A = \{(x, y): 0 < x < 4, 0 < y < 3\}$$

$$B = \{(x, y): 2 < x < 4, 2 < y < 4\}$$

Complement

Let A be any event, then the **complement** of A (denoted by A^c or $\bar{A}$) is defined by:

$$A^c = \bar{A} = \{e \mid e \text{ does not belong to } A\}$$



In problems you will recognize that you are working with:

Union if you see keywords like **or**, **either** ...

Intersection if you see the keywords like **and**, **both**, ...

Complement if you see the word **not**.

Example 2

Roll a die. Let

A = “roll an even number”

B = “roll a number less than 4”

Then: $A = \{2, 4, 6\}$, $B = \{1, 2, 3\}$

Find $A \cup B$, $A \cap B$, $\bar{A}$

$A \cup B =$

$A \cap B =$

$\bar{A} =$

Properties

- Commutative laws:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributive law)
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive law)
- $A \cup (B \cup C) = (A \cup B) \cup C$ (Associative law)
- Idempotent laws:

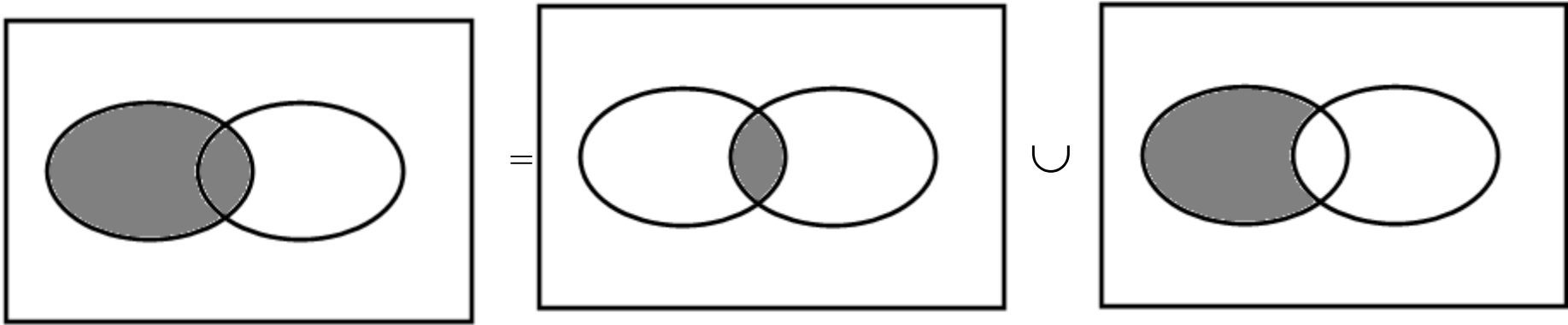
$$A \cup A = A, \quad A \cap A = A$$

- $\overline{(\overline{A})} = A$
- $\overline{A \cup B} = \overline{A} \cap \overline{B}$ (De-Morgan's Law)
- $\overline{A \cap B} = \overline{A} \cup \overline{B}$ (De-Morgan's Law)
- Inverse laws:

$$A \cup \overline{A} = S, \quad A \cap \overline{A} = \emptyset$$

Partitioning a Set

$$A = (A \cap B) \cup (A \cap \bar{B})$$



In words

The event A **occurs** if
 A **occurs** and B **occurs**

or

A **occurs** and B **doesn't occur**.

Rules involving the empty set and the sample space

$$1. \quad A \cup \phi = A$$

$$2. \quad A \cap \phi = \phi$$

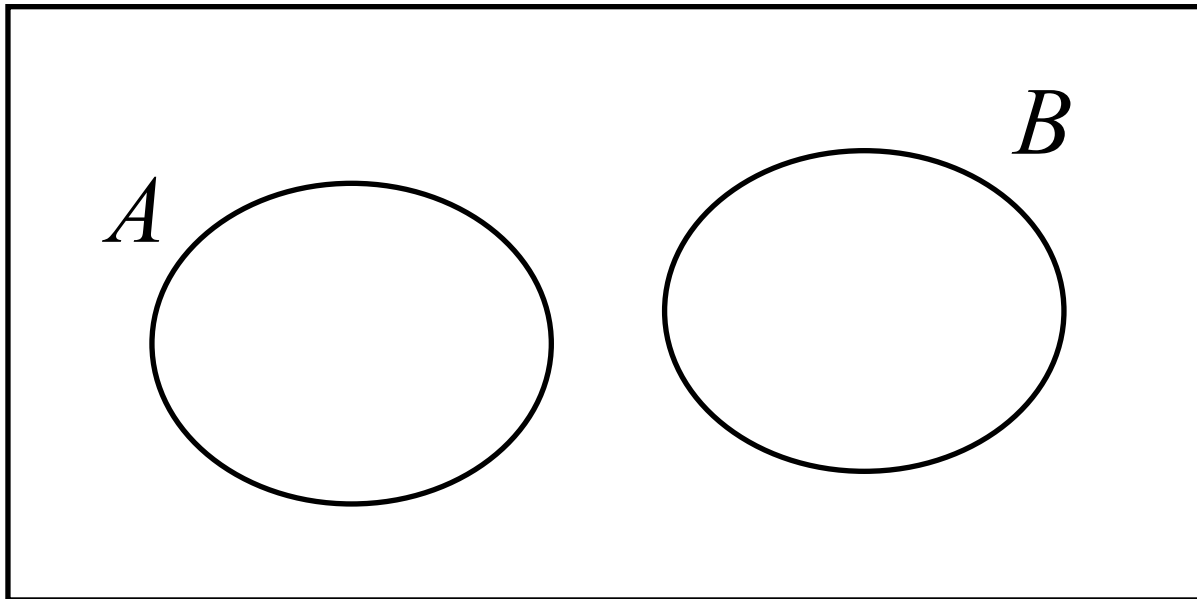
$$3. \quad A \cup S = S$$

$$4. \quad A \cap S = A$$

Definition

Two events A and B are called *mutually exclusive* or *disjoint* if:

$$A \cap B = \phi$$



Example (cars at intersection, continued):

A and C are mutually disjoint.

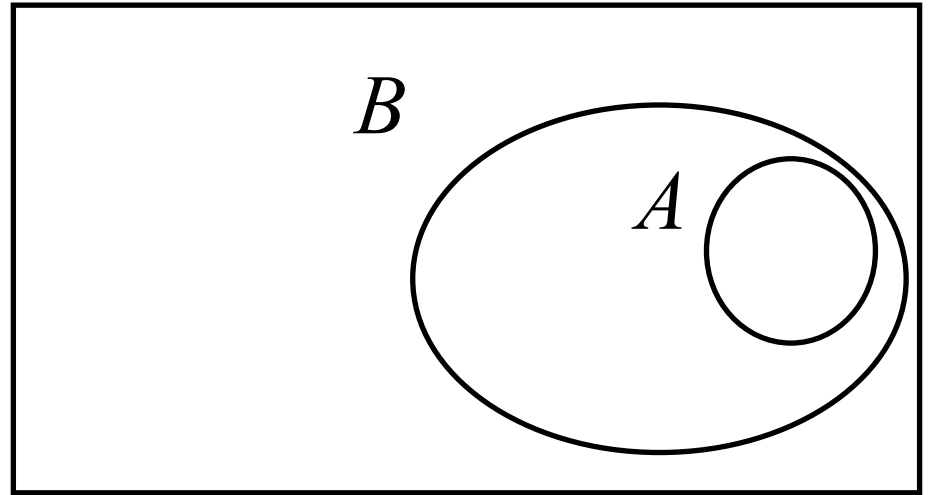
Subsets

The notation

$$A \subset B \text{ (equivalently, } B \supset A)$$

means that A is a **subset** B (B is a **superset** of A).

That is, if $e \in A$, then $e \in B$



Note:

Set inclusion fully satisfies the axioms of a partial order, but it fails the totality axiom required for a total (linear) order (for any two arbitrary sets $A, B \in \mathcal{P}(\Omega)$, either $A \subseteq B$ or $B \subseteq A$).

Counterexample: Let $\Omega = \{1, 2\}$ with subsets $A = \{1\}$ and $B = \{2\}$.

Here, $A \not\subseteq B$ and $B \not\subseteq A$.

Subsets Properties

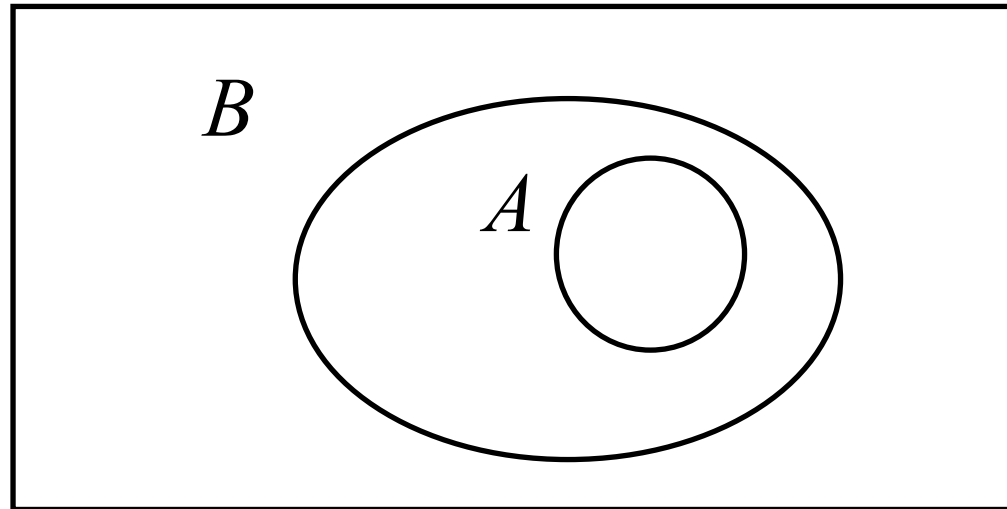
- If

$$A \subset B$$

then:

$$A \cap B = A$$

$$A \cup B = B$$



- $A \subset B$ iff $\bar{A} \supset \bar{B}$
- Time to think: How do we prove such results?

Exercise

Determine which of the following sets are subsets of which other sets.

$$A, \quad B, \quad A \cup B, \quad \bar{A} \cap B$$

Solution:

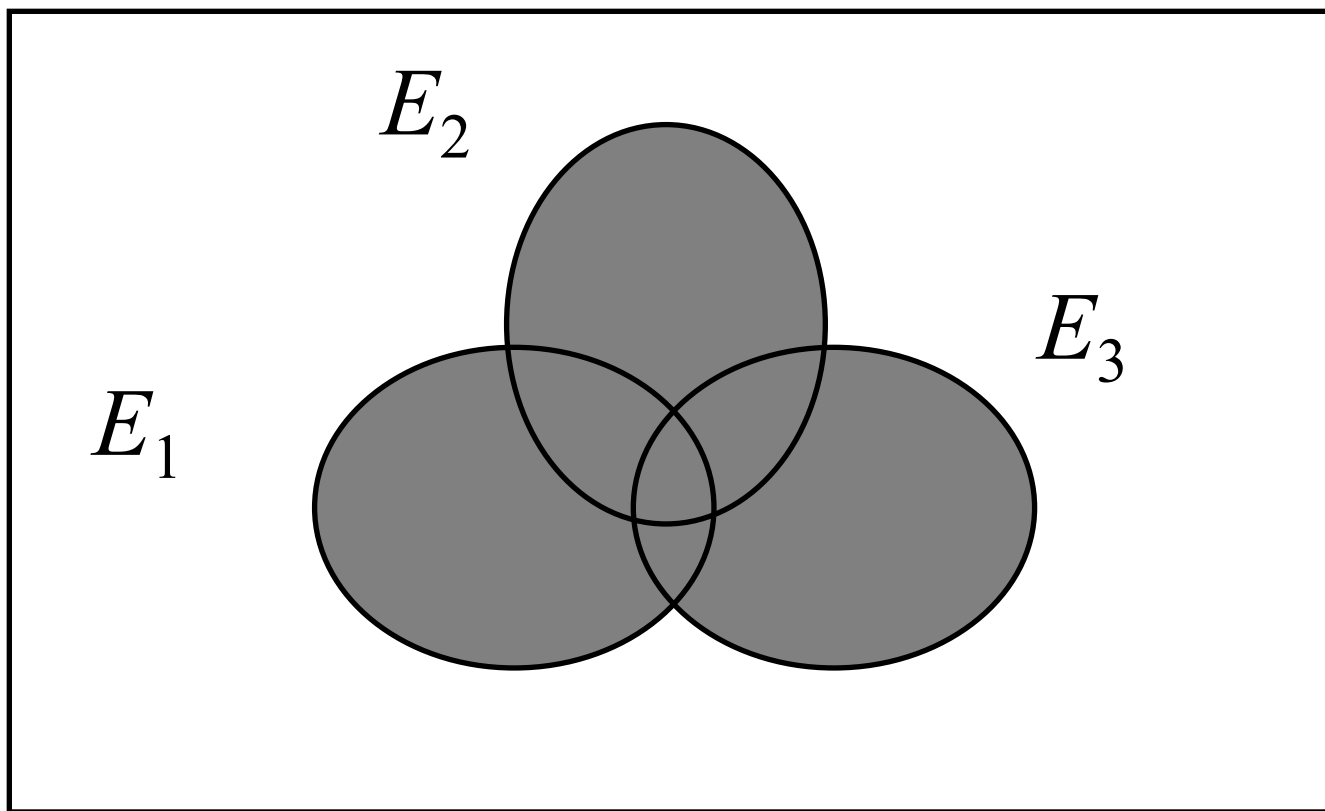
Union and Intersection

more than two events

Union:

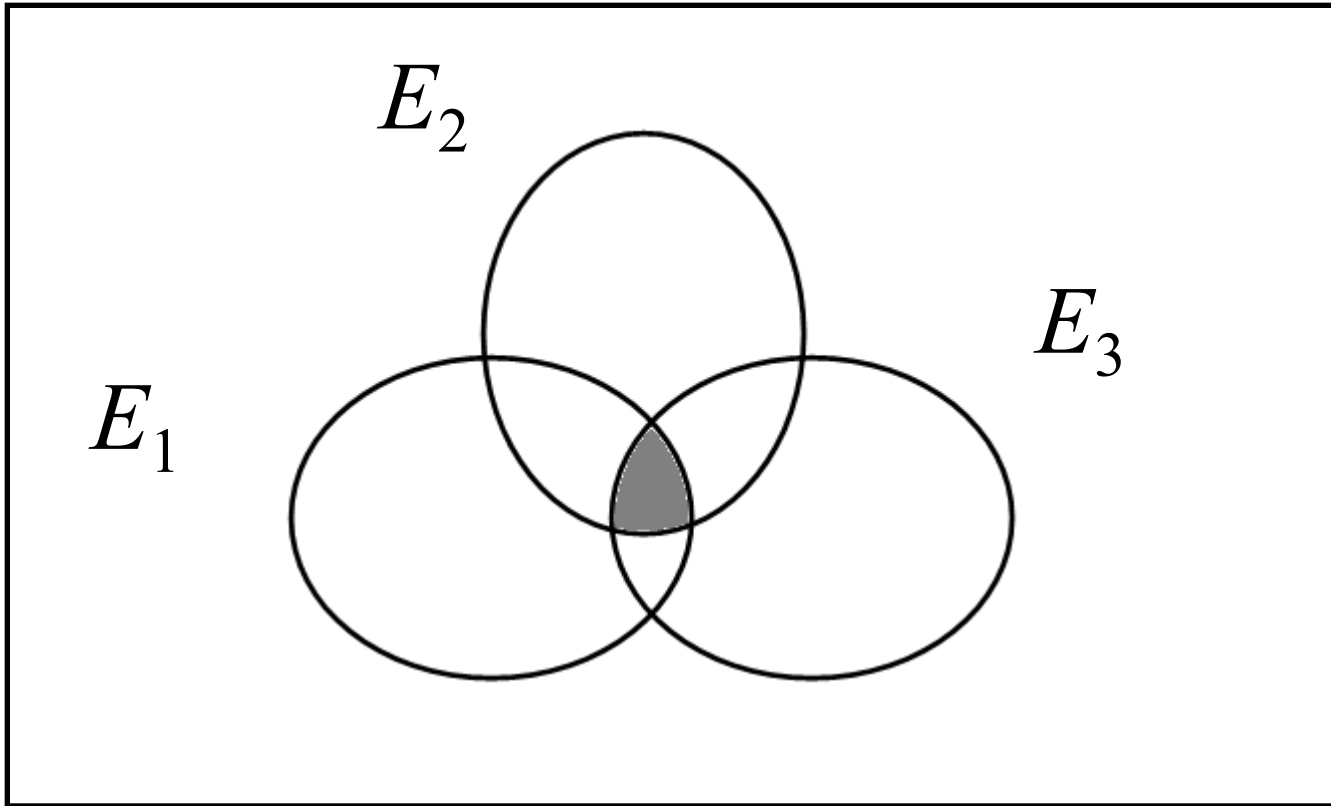
$$\bigcup_{i=1}^k E_i = E_1 \cup E_2 \cup E_3 \cup L \cup E_k$$

$$\bigcup_{i=1}^{\infty} E_i = E_1 \cup E_2 \cup E_3 \cup L$$



Intersection: $\bigcap_{i=1}^k E_i = E_1 \cap E_2 \cap E_3 \cap L \cap E_k$

$\bigcap_{i=1}^{\infty} E_i = E_1 \cap E_2 \cap E_3 \cap L$



Generalized DeMorgan's laws

For an arbitrary collection of events A_i indexed by a set I (where I can be finite, countably infinite, or uncountably infinite) within a sample space S , De Morgan's Laws generalize mathematically as follows:

- **Generalized Complement of a Union:** The complement of the union of all events is equal to the intersection of their individual complements.

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c$$

- **Generalized Complement of an Intersection:** The complement of the intersection of all events is equal to the union of their individual complements.

$$\left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c$$

Definition of Probability

Suppose we are observing a random phenomena.

Let S denote the sample space for the phenomena, the set of all possible outcomes and let the subset E of S be an event.

A **probability measure** P is defined on S by defining for each event E , $P(E)$ with the following axioms:

1. $P(E) \geq 0$, for each E .
2. $P(S) = 1$.
3. If $E_1, E_2, \dots$ are disjoint events, then:

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Simple Sample Spaces

In many examples the sample space S contains finite number, N , of outcomes. That is,

$$S = \{o_1, o_2, o_3, \dots o_N\}$$

Also, assume each of the outcomes is equally likely. Then

$$P(\{o_i\}) = \frac{1}{N}$$

Such sample spaces are known as simple or equally likely.

Then for an event E which contains m outcomes we have:

$$P(E) = \frac{m}{N} = \frac{n(E)}{n(S)} = \frac{n(E)}{N} = \frac{\text{\textit{\# of outcomes in } E}}{\text{\textit{total \# of outcomes}}}$$

Note: $n(A)$ means the number of elements of A

Example 1 (continued)

Back to example with cars:

Three cars at an intersection can turn either left (L) or right (R).

A = “Exactly one car turns right”

C = “All cars turn in the same direction”

Find $P(A)$, $P(C)$

Exercise

Roll two fair dice

Let A = “roll sum greater than or equal to 10”

Find $P(A) =$

Recap and Next Time

Many examples fall into this category

1. Finite number of outcomes
2. All outcomes are equally likely
3.
$$P(E) = \frac{n(E)}{n(S)} = \frac{n(E)}{N} = \frac{\text{\# of outcomes in } E}{\text{total \# of outcomes}}$$

Note: $n(A)$ = no. of elements of A

To handle such problems, we must be able to count. Count $n(E)$ and $n(S)$.